SG No. 59 D_{2h}^{13} $Pmmn$ [orthorhombic]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[3,4,5,6]`

Table 2: Wyckoff site: `2b`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[3,4,5,6]`

Table 3: Wyckoff site: `4c`, site symmetry: `-1`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,5]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,6]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,7]`
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[4,8]`

Table 4: Wyckoff site: `4d`, site symmetry: `-1`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,5]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,6]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,7]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,8]`

Table 5: Wyckoff site: `4e`, site symmetry: `m..`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,8]`
2	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[2,7]`
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[3,6]`
4	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[4,5]`

Table 6: Wyckoff site: `4f`, site symmetry: `.m.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z\right]$	`[2,8]`

continued ...

Table 6

No.	position	mapping
3	$\left[-x,\, \frac{3}{4},\, -z\right]$	`[3,5]`
4	$\left[x+\frac{1}{2},\, \frac{3}{4},\, -z\right]$	`[4,6]`

Table 7: Wyckoff site: `8g`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[2]`
3	$\left[-x,\, y+\frac{1}{2},\, -z\right]$	`[3]`
4	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	`[4]`
5	$[-x,\, -y,\, -z]$	`[5]`
6	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	`[6]`
7	$\left[x,\, \frac{1}{2}-y,\, z\right]$	`[7]`
8	$\left[\frac{1}{2}-x,\, y,\, z\right]$	`[8]`